THIS MARGIN

5.

$$f(x) = (3-2x)^4$$

$$f'(x) = 4(3-2x)^3 \times (-2)$$

$$f'(1) = 4(3-2(4))^3 \times (-2)$$

$$= 4(-5)^3 \times (-2)$$

$$= 1000$$

Question	Generic scheme	Illustrative scheme	Max mark
5.	$\bullet^1$ start to differentiate	$\bullet^1$ $4(3-2x)^3\ldots$	3
	$\bullet^2$ complete differentiation	$\bullet^2$ $\ldots\times(-2)$	
	$\bullet^3$ calculate rate of change	$\bullet^3$ 1000	

Notes:

1. Correct answer with no working, award 0/3.
2. Accept $4u^3\times(-2)$ where $u=3-2x$ for $\bullet^1$.
3. Where candidates evaluate $f(4)$, award 0/3, see Candidate B.
4. $\bullet^3$ is only available for evaluating expressions equivalent to $k(3-2x)^3$.

Commonly Observed Responses:

Candidate A $f'(x)=4(3-2x)^3\times(-2)$ $\bullet^1$ ✓ $\bullet^2$ ✓ $f'(x)=8(3-2x)^3$ $f'(4)=-1000$ $\bullet^3$ ✗	**Candidate B - evaluating f(x)** $f'(x)=(3-2x)^4$ $\bullet^1$ ✗ $\bullet^2$ ✗ $f'(4)=625$ $\bullet^3$ ✗
Candidate C - differentiating over two lines $4(3-2x)^3$ $\bullet^1$ ✓ $4(3-2x)^3\times 2$ $\bullet^2$ ✗ -1000 $\bullet^3$ $\boxed{\checkmark_1}$	**Candidate D - differentiating over two lines** $4(3-2x)^3$ $\bullet^1$ ✓ $4(3-2x)^3\times -2$ $\bullet^2$ ^ 1000 $\bullet^3$ $\boxed{\checkmark_1}$
Candidate E - insufficient evidence for mark 1 $f'(x)=8(3-2x)^3$ $\bullet^1$ ✗ $\bullet^2$ ✗ $f'(4)=-1000$ $\bullet^3$ $\boxed{\checkmark_1}$	**Candidate F** $4(3-2x)^3$ $\bullet^1$ ✓ $\bullet^2$ ^ -500 $\bullet^3$ $\boxed{\checkmark_1}$